

A Totally Real Polynomial for $\text{PSL}_2(\mathbf{F}_{23})$

The following monic polynomial is irreducible over $\mathbf{Q}$, has exactly 24 real roots, and has splitting field with Galois group $\text{PSL}_2(\mathbf{F}_{23})$:

$$\begin{aligned} f_{23}(x) = & x^{24} - 9x^{23} \\ & - 34023233x^{22} \\ & + 23909574905x^{21} \\ & + 378784186692003x^{20} \\ & - 416111841180211061x^{19} \\ & - 1758976142457884739419x^{18} \\ & + 2498354112494426078941203x^{17} \\ & + 3443986147522394798725590240x^{16} \\ & - 6126664469966568538343243665148x^{15} \\ & - 2871902716629301402769031406507036x^{14} \\ & + 7113714939144568181345193644751461856x^{13} \\ & + 953908328438603735842006274581705046890x^{12} \\ & - 4420312281698269204076091946283398220987868x^{11} \\ & + 3558840808355835785638783068002842746072176x^{10} \\ & + 1531303302156485034582579759710699903755356002084x^9 \\ & - 73057343599496156146996562517802101572094242427693x^8 \\ & - 286283268829668024007089337603418027154834506144718859x^7 \\ & + 15273515412889515974820218228067160899232094041453394119x^6 \\ & + 25662472416605616304774572851371028375104520635826997326115x^5 \\ & - 1208880635983675443731875655352952561003817465802472714911667x^4 \\ & - 920551755968570931232226909091292226795924891723769872837511125x^3 \\ & + 24814780529781344306979190165045565482857618848707583326292941867x^2 \\ & + 9200519549011006510235576352293488802205345340057882055051922762255x \\ & - 101992260865864218313977648508190555361360255037688677072099502721850. \end{aligned}$$